mass_properties.py | lines 157-169

```
    inertia = np.zeros((3, 3))
    inertia_rate = np.zeros((3, 3))
    for c in components:
        distance = c.center_axial_m - center
        distance_rate = c.center_rate_mps - center_rate
        parallel = np.diag([0.0, distance**2, distance**2])
        parallel_rate = np.diag([0.0] + [2 * distance * distance_rate] * 2)
        inertia += c.inertia_at_center_kgm2 + c.mass_kg * parallel
        inertia_rate += (
            c.inertia_rate_at_center_kgm2ps
            + c.mass_rate_kgps * parallel
            + c.mass_kg * parallel_rate
        )
```